INDEPENDENCE OF THE LOCALLY CONSTRUCTED BIVARIANT CLASS FROM THE CHOICE OF THE COMPLEMENTARY CLOSED SUBSCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FAH (sources/stacks-chow/pinned-original-source.tex line 9752).

Allowed standard prerequisites (editorial).

- The local construction of P'_p(E_2) and c'_p(E_2) and its two hypotheses. Same chapter, Lemma labelled lemma-silly (source line 9695), whose hypotheses the statement inherits. Fixes the situation X = X_1 union X_2, the closed subscheme E_2 and the condition on the restriction of E_2 to X_1 cap X_2, whose choice is what this lemma shows to be immaterial.
- Decomposition of a cycle class along a cover by two closed subschemes and bivariant capping. Same chapter, chow.tex, the cycle group Z_k(X) and the bivariant operation used at proof lines 9764-9769. Supplies that alpha = i_*beta + i_{2,*}alpha_2 with beta supported on the intersection and alpha_2 on X_2, which is the step that makes the two constructions agree.

Source reference. Original lemma 0F9H: sources/stacks-chow/pinned-original-source.tex, source lines 9695–9695. Same-chapter pointer into the author environment lemma at source line 9695; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Lemma 0F9H the bivariant class $P'_p(E_2)$, resp. $c'_p(E_2)$ in $A^p(X_2 \to X)$ does not depend on the choice of X_1.*

Proof. Suppose that $X'_1 \subset X$ is another closed subscheme such that $X = X'_1 \cup X_2$ set theoretically and the restriction $E_2|_{X'_1 \cap X_2}$ is zero, resp. isomorphic to a finite locally free $\mathcal{O}_{X'_1 \cap X_2}$-module of rank $< p$ sitting in cohomological degree 0. Then $X = (X_1 \cap X'_1) \cup X_2$. Hence we can write any element $\alpha \in \mathrm{CH}_k(X)$ as $i_*\beta + i_{2,*}\alpha_2$ with $\alpha_2 \in \mathrm{CH}_k(X'_2)$ and $\beta \in \mathrm{CH}_k(X_1 \cap X'_1)$. Thus it is clear that $P'_p(E_2) \cap \alpha =$

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$P_p(E_2) \cap \alpha_2 \in \mathrm{CH}_{k-p}(X_2)$, resp. $c'_p(E_2) \cap \alpha = c_p(E_2) \cap \alpha_2 \in \mathrm{CH}_{k-p}(X_2)$, is independent of whether we use X_1 or X'_1. Similarly after any base change. $\square$